

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/323580470>

Optimal adaptation equivalent factor of energy management strategy for plug-in CVT HEV

Article in *Proceedings of the Institution of Mechanical Engineers Part D Journal of Automobile Engineering* · March 2018

DOI: 10.1177/0954407018755612

CITATIONS

22

READS

1,696

4 authors, including:



Xinyou Lin

Fuzhou University

57 PUBLICATIONS 979 CITATIONS

[SEE PROFILE](#)



Mo Liping

Xiamen University

12 PUBLICATIONS 191 CITATIONS

[SEE PROFILE](#)

Optimal adaptation equivalent factor of energy management strategy for plug-in CVT HEV

Proc IMechE Part D:
J Automobile Engineering
2019, Vol. 233(4) 877–889
© IMechE 2018
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/0954407018755612
journals.sagepub.com/home/pid
 SAGE

Xinyou Lin¹, Qigao Feng¹, Liping Mo¹ and Hailin Li²

Abstract

This study presents an adaptive energy management control strategy developed by optimally adjusting the equivalent factor (EF) in real-time based on driving pattern recognition (DPR), to guarantee the plug-in hybrid electric vehicle (PHEV) can adapt to various driving cycles and different expected trip distances and to further improve the fuel economy performance. First, the optimization model for the EF with the battery state of charge (SOC) and trip distance were developed based on the equivalent consumption minimization strategy (ECMS). Furthermore, a methodology of extracting the globally optimal EF model from genetic algorithm (GA) solution is proposed for the design of the EF adaptation strategy. The EF as the function of trip distances and SOC in various driving cycles is expressed in the form of map that can be applied directly in the corresponding driving cycle. Finally, the algorithm of DPR based on learning vector quantization (LVQ) is established to identify the driving mode and update the optimal EF. Simulation and hardware-in-loop experiments are conducted on synthesis driving cycles to validate the proposed strategy. The results indicate that the optimal adaption EF control strategy will be able to adapt to different expected trip distances and improve the fuel economy performance by up to 13.8% compared to the ECMS with constant EF.

Keywords

Driving pattern recognition, energy management system, equivalent factor, genetic algorithm, plug-in hybrid electric vehicle

Date received: 1 January 2017; accepted: 2 January 2018

Introduction

With up-front costs and range anxiety often cited as detractors for fully electric vehicles (EV), plug-in hybrid electric vehicles (PHEV) offer a potentially attractive alternative for consumers and fleets seeking lower running costs and tailpipe emissions.¹ Compared with the traditional hybrid electric vehicles (HEV), PHEV enable more fuel economy improvements due to a larger capacity battery pack that can be recharged from the electric grid.

Different from HEV, the PHEV can work in different modes which are defined as EV mode, charge depleting (CD) mode and charge sustaining (CS) mode according to the battery state of charge (SOC) value and the driving demand power.² As it shown in Figure 1, the SOC_{min} and SOC_{max} represent the lower and upper limit of the battery SOC. During EV mode, the electric motor (EM) will be the only drive power to consume the battery power. During CS mode, when the SOC reaches a certain value approximately, the

engine will become the primary drive power to prevent the battery from over discharge and the variation of SOC will be maintained within a small range. The blended CD mode, which is between the EV mode and CS mode, will consume battery power more gradually. Driving mode is of great importance in selecting the energy management strategy. Since driving information in real world operation is difficult to obtain in real time, PHEVs might always be bound to work in the CD mode for an approximate optimal SOC trajectory.³

¹College of Mechanical Engineering and Automation, Fuzhou University, China

²Department of Mechanical and Aerospace Engineering, West Virginia University, USA

Corresponding author:

Xinyou Lin, College of Mechanical Engineering and Automation, Fuzhou University, Fuzhou 350000, China.
Email: Linxinyou@fzu.edu.cn

The excellent fuel economy of PHEV depends not only on its extended energy storage, but also on the energy management.⁴ Typically, the energy management strategies can be divided into two categories, namely the rule based and optimization based method.^{5,6} The objective of the optimization based energy management strategy is to acquire an optimal control variable law within feasible constraints. Optimization based strategies can be further classified into real-time optimization and global optimization.

An optimal SOC trajectory is a global minimization problem of the total energy consumption, which forms a challenging technology for PHEV control.⁷ The global control generally includes dynamic programming (DP) and a genetic algorithm (GA).⁸ DP is a kind of exhaustive search algorithm. Through splitting of the complex problem into numbers of sub-problems, DP will give the optimal control which usually is regarded as the benchmark of design.⁹ Peng et al.¹⁰ proposed a recalibration method to improve the fuel economy of the rule-based energy management through the results calculated by a DP algorithm. GA emulates the evolution in nature. Typically, the kernel processes of GA consist of selection, encoding, mutation and crossover. The value of fitness function (generally the fuel consumption for PHEVs strategies) decides which kind of gene will be selected to generate the next generation. It is extremely effective to implement the GA to acquire the global optimal solution. Besides, the inherent parallelism of GA makes it easy to speed up the computation. This algorithm has been used to improve fuel economy effectively.^{11,12} However, the global optimization always requires a control system with large quantity of computer memory, as well as the details of the trip information, which usually limits its utility to off-line optimization.

The model predictive control (MPC) and the stochastic dynamic programming (SDP) are frequently used in real-time optimization control,^{13–15} and are expected to get close to the global optimal solution utilizing algorithms of the energy management strategy. MPC will minimize a series of cost functions over receding time horizons to maintain the computational burden within an acceptable range.¹⁶ In Zhao and Wu,¹⁷ a Markov chain model is developed to predict the future demand power, and then the SDP is used to minimize the expectation of fuel consumption in a limited and suitable horizon. However, the prediction accuracy and predicted horizons are the problems remaining to be solved, which will affect the performance of the MPC significantly in an actual application.¹⁸

The ECMS simplifies the global dynamic optimization problem to an equivalent local (instantaneous) optimization question based on Pontryagin's minimum principle (PMP) which reduces the computational complexity and fit for a real-time controller.¹⁴ The underlying principle of the ECMS approach is that an equivalent factor (EF) is assigned to convert the electrical energy into a certain quantity of fuel.^{19,20} The value

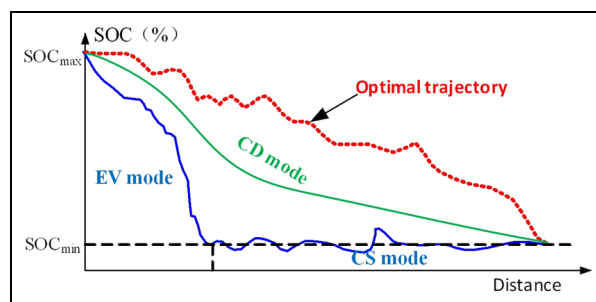


Figure 1. The SOC trajectory in EV, CD, CS modes.

of SOC plays an important role in the ECMS energy control strategy.²¹ For conventional hybrid EVs, the key idea of ECMS is to determine EF in both charge and discharge.²² PHEV will work with a wider SOC range (typically 0.2–0.9) compared with the HEVs, which increases the complexity to estimate the optimal EF. Besides, the optimal EF depends significantly on driving cycles because the SOC trajectory is varying from different driving patterns. Adaptive ECMS (A-ECMS) is a practical way to tune the EF online to improve the fuel economy. The A-ECMS is classified into: driving cycle prediction adaptation, driving pattern adaptation and SOC adaptation.²³ Driving cycle prediction adaptation uses the historical trip information to predict future vehicle speeds and accelerations, then with the predefined optimal EF in the predicted cycle, the EF will be tuned online. In Tianheng et al.,²⁴ an artificial neural network (ANN) was developed to predict the demand power that would be translated to the reference SOC trajectory, then, A-ECMS was used to follow the reference SOC, and the simulation result showed the fuel economy is close to that of DP.

With the development of GPS and GIS, some of the trip information can be known in advance including the trip distance in a specific cycle. The trip distance is a significant characteristic for PHEV due to the limited all electrical range (AER). When the remaining trip distance is less than AER, the vehicle should drive in pure electric mode to consume battery power as much as possible to improve the fuel economy. Otherwise, to obtain a better fuel economy, the electrical energy should be consumed gradually and reach the SOC minimum limit at the terminal of trip. With the trip distance information, A-ECMS is proposed to improve the fuel economy.²⁵ Moreover, in Zhang and Vahidi²⁶ the EF as the function of SOC and trip distance calculated by DP and stored in the form of maps to tune the EF in real-time.

The main contributions of this paper are as follows. The mathematical function of EF with SOC and distance is developed to adjust the control strategy depending on the various trip destinations. Furthermore, a method is proposed for extracting the globally optimal EF map from GA and the Urban Dynamometer Driving Schedule (UDDS), New York City Cycle (NYCC) and Highway Fuel Economy Test (HWFET) driving cycles are used to demonstrate the method. This

Table 1. PHEV powertrain specification.

	Parameter	Value	Unit
Engine	Max power P_{emax}	63	kw
	Max torque T_{emax}	150@4000 rpm	N · m
	Speed range N_e	800–6000	rpm
Motor	Normal power P_m	32	kw
	Max torque T_{mmax}	140	N · m
	Speed range N_m	0–6000	rpm
ISG	Normal power P_{ISG}	25	kw
	Max torque T_{ISGmax}	110	N · m
	Speed range N_m	0–6000	rpm
Battery	Capacity Q_0	51	A · h
	Normal voltage U_0	320	V
CVT	Ratio range i_{cvt}	0.422–2.432	–

method can provide valuable insights into how to adjust EF for achieving a global optimal performance as a benchmark for online ECMS adaptation. Finally, the driving pattern recognition (DPR) algorithm is developed to adjust the EF online for adapting the various driving cycles. Based on this approach, a novel adaptive energy management control strategy is formulated.

The remainder of the paper is structured as follows: Section 2 formulates the EF with a trip distance optimization mathematical model and extracts the optimal EF map for various driving cycles. Section 3 develops the driving cycle recognition based learning vector quantization (LVQ). Section 4 investigates the A-ECMS Energy Management strategy, which is established based on the optimal map of EF combined with driving pattern recognition. The simulation and hardware in the loop (HIL) experiment are carried out in Section 5. Finally, the conclusions are presented in Section 6.

Model of the PHEV system

Powertrain of a series-parallel PHEV

The prototype of this study is a plug-in series-parallel HEV with continuously variable transmission (CVT).

The powertrain is shown in Figure 2, including a gasoline engine, an EM, an integrated starter generator (ISG) and a battery pack. The vehicle can work in both series and parallel mode by controlling the state of the clutch. The powertrain parameters of the vehicle are listed in Table 1.

System modeling

Vehicle dynamic model. The longitudinal dynamics of the vehicle are implemented to evaluate the fuel economy performance. The equation can be written from the equilibrium of forces as follows

$$\begin{cases} F = m_v g(\sin \theta + f_r \cos \theta) + \frac{1}{2} C_d A v_v^2 + \delta m_v \frac{dv}{dt} \\ F = F_{out} + F_{brake} \end{cases} \quad (1)$$

The power balance can be written

$$\begin{aligned} P_{out} &= \left(m_{veh} g(\sin \theta + f_r \cos \theta) + \frac{1}{2} C_d A v_{veh}^2 + \delta m_{veh} \frac{dv}{dt} \right) \cdot v \end{aligned} \quad (2)$$

Furthermore P_{out} is expressed as follows

$$P_{out} = i_0 i_{cvt} \eta_t (P_{eng} + P_{EM} + P_{ISG}) \quad (3)$$

The total power at the wheel, which is the sum of the powertrain output power P_{out} and braking power P_{bra} (including regenerative braking power), should be equal to the driver demand power P_{demand} to meet the driving constraints.

$$P_{demand} = P_{out} + P_{bra} \quad (4)$$

Engine model. To optimize the fuel economy performance of the vehicle, a static model of the engine is used based on the fuel consumption per kilowatt-hour data from the experiment shown in Figure 3.

The engine fuel consumption can be described by

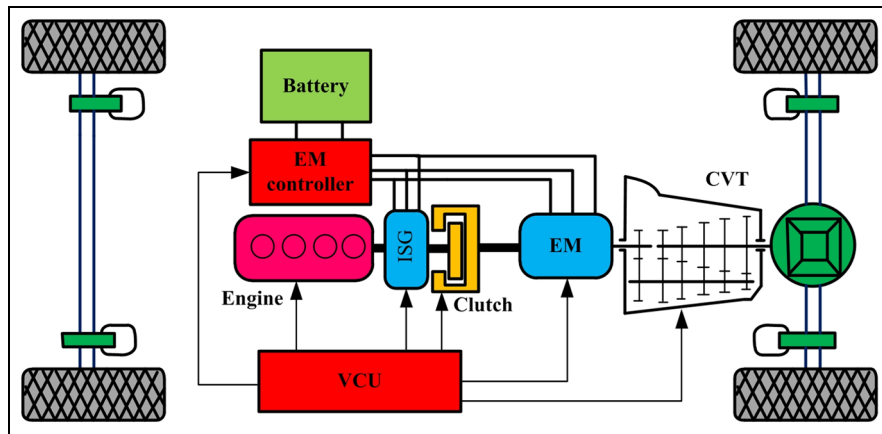


Figure 2. Powertrain of a series-parallel plug-in hybrid electric vehicle. VCU is the vehicle control unit.

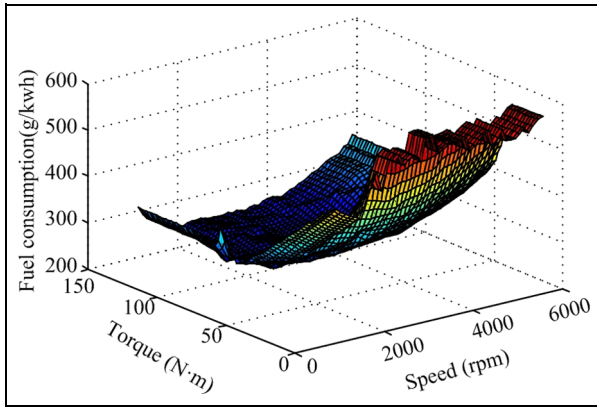


Figure 3. Map of the engine fuel consumption in feasible operation conditions.

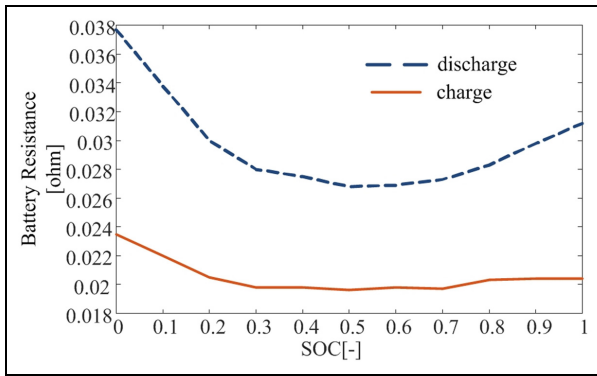


Figure 4. The resistance at the charging and discharging state.

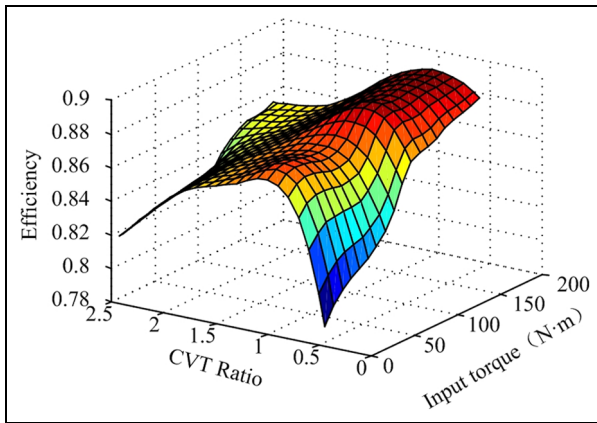


Figure 5. Efficiency of CVT with respect to gear ratio and input torque.

$$\begin{cases} \dot{m}_{eng} = \frac{1}{3.6 \times 10^6} b_e \cdot T_{eng} \cdot \omega_{eng} \\ \quad = \frac{1}{3.6 \times 10^6} b_e \cdot P_{eng}(\omega_{eng}) \\ m_{eng} = \int_{t_0}^{t_f} \dot{m}_{eng} dt \end{cases} \quad (5)$$

The engine efficiency can be defined as

$$\eta_e(T_{eng}, \omega_{eng}) = \frac{T_{eng} \cdot \omega_{eng}}{\dot{m}_{eng} Q_{lhv}} \quad (6)$$

Electric motors model. The efficiency of EM and ISG can be obtained by means of an efficiency map based on the experiment data, and expressed as follows

$$\eta_{m, isg} = f(T_m, \omega_m) \quad (7)$$

Battery model. The battery SOC is set in the range of 0.3–0.9 to protect the battery from overcharge and over discharge. Since the power flow is the research focus, the impacts of temperature change and battery degradations are not considered and the battery and traction motor are assumed as a typical equivalent circuit model.

The SOC is calculated by

$$\dot{SOC} = -(I_{batt}/Q_0) \quad (8)$$

$$P_{batt} = V_{oc} \cdot I_{batt} - I_{batt}^2 R_{batt} \quad (9)$$

It is important to note here, the resistance of the battery as a function of SOC at charging and discharging states are shown in Figure 4.

The battery power in PHEV contains the power supply for EM and power generated by ISG. The terminal battery power is formulated as

$$P_{batt} = \frac{P_{EM}}{(\eta_{EM})^{k_{EM}}} + \frac{P_{ISG}}{(\eta_{ISG})^{k_{ISG}}} \quad (10)$$

$$k_i = \begin{cases} 1, & P_i > 0 \\ -1, & P_i < 0 \end{cases}, \text{ for } i = \{EM, ISG\}$$

CVT model. In this research, the CVT provides the speed ratio from 0.422 to 2.432 and can adjust operation points for the engine and EM via changing gear ratios between the desired limits.²⁷

To meet the requirements of fuel economy evaluation, the efficiency of the CVT with the relationship of the speed ratio and driving torque is contained in a map which can be used directly in the evaluation model, as shown in Figure 5. The efficiency of CVT can be expressed as follows

$$\eta_{cvt} = f(i_{cvt}, T_{cvt}) \quad (11)$$

Problem statement for standard ECMS

For PHEVs, the battery is requested to work under a different mode according to the different driving conditions. The object of the control strategy is to minimize the fuel consumption over the global cycle.

In this study, $SOC(t)$ is set as the state variable $x(t)$, battery power P_{batt} and engine speed ω_{eng} are set as control variables $u(t)$. The cost function is as follows

$$J(x, u) = \int_{t_0}^{t_f} \dot{m}_{equ}(u) dt, \quad \dot{x}(t) = f(x, u) \quad (12)$$

Subject to the constraints as follows

$$\begin{cases} P_{batt_min} \leq P_{batt} \leq P_{batt_max} \\ SOC_{min} \leq SOC \leq SOC_{max} \\ P_{eng_min} \leq P_{eng} \leq P_{eng_max} \\ \omega_{eng_min} \leq \omega_{eng} \leq \omega_{eng_max} \end{cases} \quad (13)$$

Where the $\dot{m}_{equ}$ is the instantaneous equivalent fuel consumption, which contains the battery equivalent consumption and the real fuel consumption from the engine at each sample time, and it is expressed as follows

$$\dot{m}_{equ} = \dot{m}_{eng} + \dot{m}_{eb} \quad (14)$$

According to the expression (5), an engine which consumes real fuel and for the unified form the expression (5) is rewritten as follows

$$\dot{m}_{eng} = \frac{1}{3.6 \times 10^6} P_{eng} \cdot b_e \quad (15)$$

Based on the idea of ECMS, the battery is regarded as the energy buffer. When the battery power is positive at the present time, it implies the battery will be recharged in the future which leads to extra fuel consumption. When the battery power is negative, the stored electrical energy will help save fuel at a certain time in the future. Therefore, the battery equivalent consumption is defined by the following expression

$$\dot{m}_{eb} = \begin{cases} S \frac{P_{batt}}{\eta_{chg} Q_{thv}}, & P_{batt} < 0 \\ S \frac{P_{batt} \eta_{dis}}{Q_{thv}}, & P_{batt} > 0 \end{cases} \quad (16)$$

Where η_{chg} and η_{dis} are the charging efficiency and discharging efficiency, respectively.

$$S = \begin{cases} \frac{\bar{\eta}_{isg} \bar{\eta}_{dis}}{\eta_e} P_{batt}(t) < 0 \\ \frac{1}{\eta_m} P_{batt}(t) \geq 0 \end{cases} \quad (17)$$

Where S is the EF (for more detail, please refer to the previously published literature^{28–30}) $\bar{\eta}_{isg}$, $\bar{\eta}_{dis}$, $\bar{\eta}_e$ and $\bar{\eta}_m$ are the mean efficiencies of ISG, battery, engine and traction motor, respectively.

On the other hand, when considering the regenerative power, in this paper, the mean regenerative power with a time interval every 120 s is implemented in the control algorithm (as shown in the Figure 6). The regenerative power is denoted by $P_r(t)$; so the battery equivalent fuel will be redefined as

$$\dot{m}_{eb} = \begin{cases} S \frac{P_{batt}(t) + P_r(t)}{\eta_{dis} Q_{thv}}, & P_{batt}(t) > 0 \\ S \frac{P_{batt}(t) \eta_{chg}}{Q_{thv}}, & P_{batt}(t) < 0 \end{cases} \quad (18)$$

Based on the above definition, for instance, the process to solve the control variable is expressed as

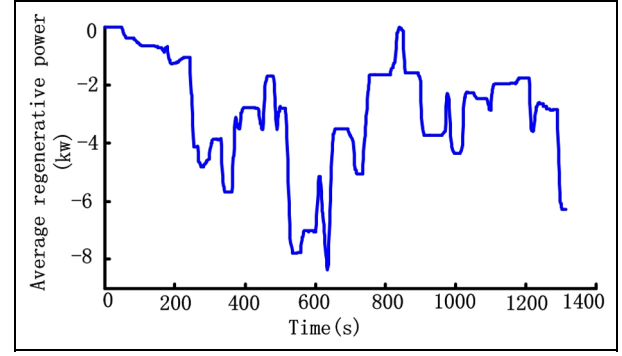


Figure 6. The mean regenerative power in 120 s.

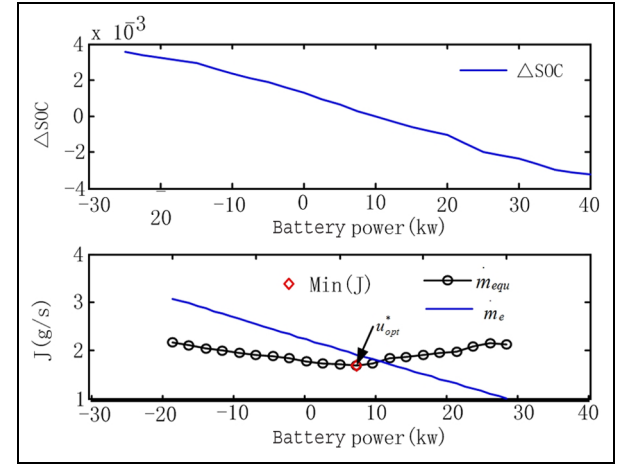


Figure 7. The process of solving the optimal battery power based ECMS.

follows. First, the accessible battery power range $[P_{batt_min}, P_{batt_max}]$ is discretized by interval ΔP_{batt} into a series of control candidates; then the instantaneous equivalent fuel consumption $\dot{m}_{equ}$ is calculated under the system constraints corresponding to each control candidate. Finally, the optimal battery power which corresponding to the minimum $\dot{m}_{equ}$ is selected, as Figure 7 shows.

The mathematical model of EF with distance

As the previously described, the fuel economy of the control strategy depends strongly on the initial SOC and the trip distance. To consider both the SOC and trip distance, the EF mathematical model is demonstrated in this section.

The core idea of the present strategy is to find the optimal control variable during the cycle to minimize the equivalent fuel consumption in (12) by adapting the algorithm to different SOC and trip distance. It is worth noting the relationship between EF and the SOC value. According to the features of PHEV, the original EF as the function of SOC is defined based the polynomial fitting as shown in Figure 8.

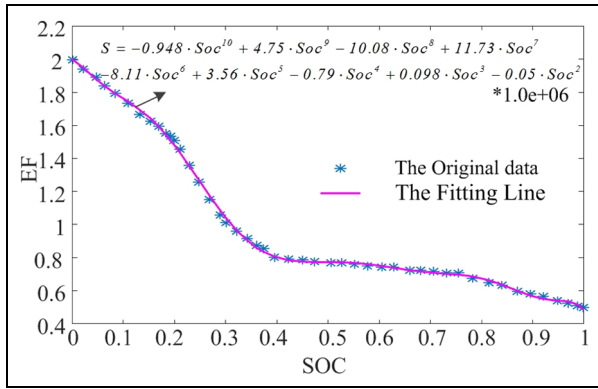


Figure 8. The relationship between EF and SOC.

The penalty factor of EF is a function of SOC, this stands for the mean when the SOC changes, and the cost of the electric and fuel will be changed. To be implemented in the real-time control, the fitting line is used as a look-up table to update the EF. The EF is noted as S , and the relationship of the S and SOC can be expressed as

$$K(Soc) = a \cdot Soc^{10} + b \cdot Soc^9 + c \cdot Soc^8 + d \cdot Soc^7 + e \cdot Soc^6 + f \cdot Soc^5 + g \cdot Soc^4 + h \cdot Soc^3 + i \cdot Soc^2 + j \cdot Soc + k \quad (19)$$

Where the a, b, c, d, k are the coefficients of the polynomial and can be estimated by using the curve-fitting method, and the specific value is $[-0.948, 4.755, -10.086, 11.736, -8.111, 3.359, -0.798, 0.098, -0.005, 0, 0] \times 1.0e + 06$.

The basic principles are described briefly. Whenever the electric energy is sufficient, that is the SOC value is high relatively, the price of battery power is considered to cost “cheap” and the EF is tuned into a smaller value at the same time in order to control the vehicle to favor battery power. Under the opposite condition, the electric energy becomes “expensive,” in which fuel energy will be preferred. In order to maintain the SOC within a feasible range, a penalty function is presented for the HEV charge-sustaining mode to control the EF.³¹ The proposed optimization model of EF considering the SOC is expressed as follows

$$S_{opt}(Soc, t) = K^* S(Soc) \quad (20)$$

where K is the penalty factor.

In order to correlate the parameter of the trip distance, the optimal reference SOC is assumed in advance and the reference SOC trajectory can be expressed as follows³²

$$SOC_{ref}(t) = SOC_{t_0} - \frac{D(t)}{\bar{D}} (SOC_{t_0} - SOC_{t_f}) \quad (21)$$

where $SOC_{ref}(t)$ is the reference SOC, SOC_{t_0} is the SOC at the beginning of the cycle, SOC_{t_f} is the SOC at the

end of the cycle, $D(t)$ is the driving distance, $\bar{D}$ is the total trip distance.

Finally, in order to estimate the EF with the knowledge of the trip distance and Soc , the Soc in expression (20) is replaced by the $Soc_{ref}(t)$ in expression (21) and the model of EF with distance is rewritten as follows

$$S_d(SOC_{int}, D(t), t) = K^* S[Soc_{ref}(t)] \quad (22)$$

Where S_d denotes the EF with distance. The $SOC_{ref}(t)$ will be calculated in a different trip distance $\bar{D}$ and Soc , and then the EF will be calculated by (22).

Extraction of optimal EF map for various driving cycles

The model-based optimization problem is formulated to obtain the optimal EF in the case of different initial SOC values and trip distances. With expression (19) the EF penalty factor K is assigned as the design variables in this problem. The total fuel consumption and SOC trajectory during the simulation are sensitive to the penalty factor K . The designed variable will be validated with the above ECMS strategy in a PHEV model developed in Simulink. The objective function is the overall fuel consumption for the target driving cycle. In this research, the UDDS, NYCC and HWFET driving cycles are selected for the case study. The cost function of the penalty factor K is represented as the equation (12), and the simulation should satisfy the constraints shown in equation (13).

$$\begin{cases} k_{opt} = \operatorname{argmin}_{\{k\}} [\int_{t_0}^{t_f} \dot{m}_{equ} dt] \\ S_{opt}(SOC_{int}, D(t), t) = K_{opt}^* S[Soc_{ref}(t)] \end{cases} \quad (23)$$

The process of GA is summarized in Figure 9. The Simulink GA function is used to implement the problem. First, the initial parameters are filled and the fitness function is the cost function in (20), the initial population is 100, the maximum generation is 80, the number of elite is 10 and the crossover rate is 0.4. Second, the fitness function is calculated in the Simulink model. Then, the fitness is compared to decide the selected individuals. After that, the crossover and variation are implemented. Finally, the process is completed with the predefined condition which can be the maximum generations in this problem.

For the robust and efficiency computation, the GA is implemented to optimize this nonlinear problem offline. The optimization result will be utilized to tune the EF online. The GA obtains the optimal results through a fitness-based process, where the optimal solution will be selected to generate the next generation. Figure 10 shows the optimization results calculated for the UDDS cycle and the optimal penalty factor is 3.182.

Three types of driving cycles, UDDS, NYCC and HWFET are selected as representative. These typical driving cycles are employed to develop the optimal adaptation EF. For a given trip distance, the battery

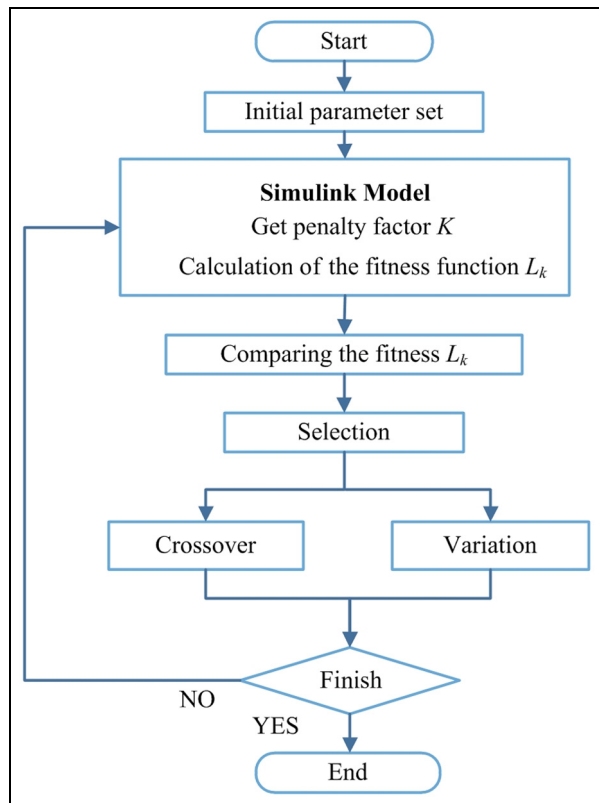


Figure 9. The optimization process of EF based on GA.

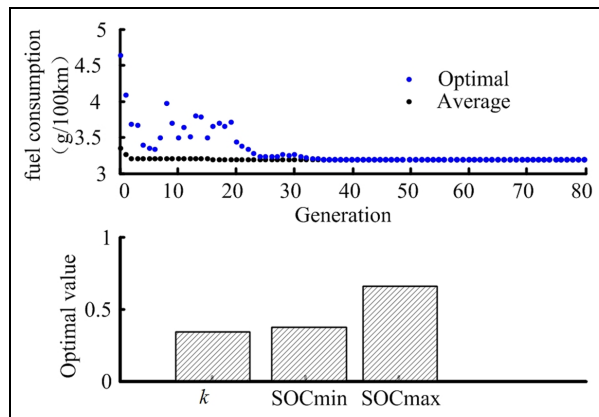


Figure 10. The optimization result of GA.

SOC, and iterative optimization process using the equation (21) is carried out for estimating optimal EF penalty factor K . Then, the calculated data is stored in the ECMS database as a map function of three variables. They are trip distance, battery SOC and optimal EF, as Figure 11 shows.

As in the previous section, the GA optimal solutions are implemented to make the inverse operation to extract the optimal EF. In the ECMS database, for a given trip distance and battery SOC, the optimal engine power and EF also have a one-to-one relationship. As a final step of the numerical (map-based) approach, the optimal EF corresponding to the optimal engine power

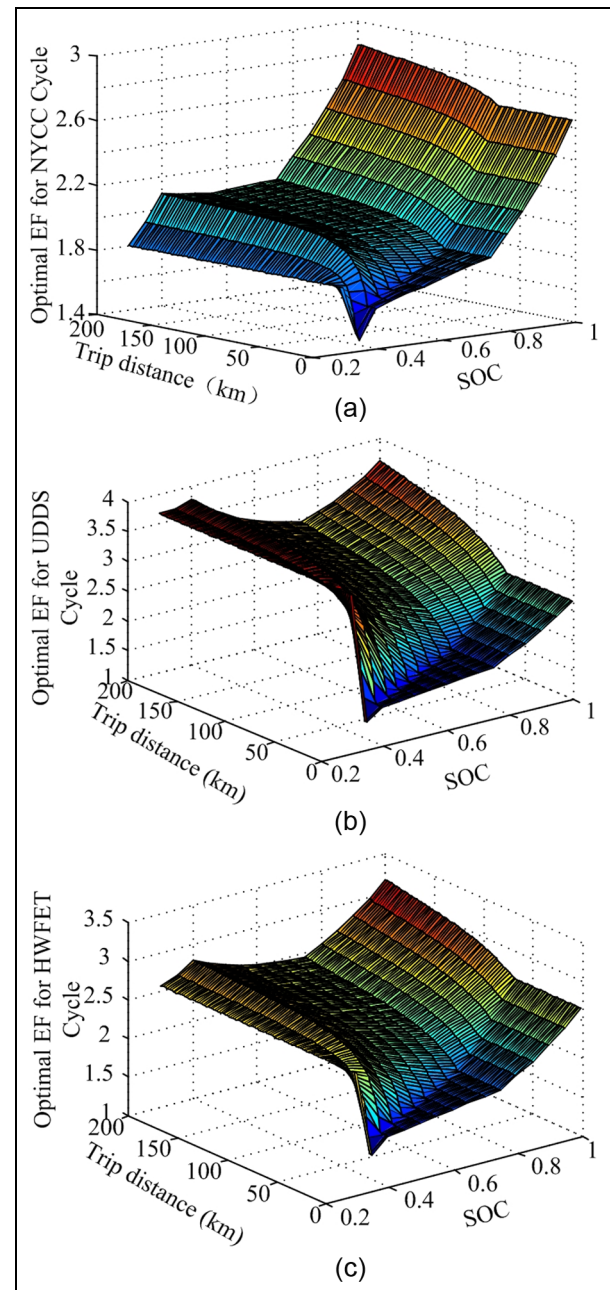


Figure 11. ECMS database: the optimal EF as a function of the battery SOC and trip distance for three driving cycles: (a) optimal EF for NYCC cycle; (b) optimal EF for UDSS cycle; (c) optimal EF for HWFET cycle.

acquired from GA can be determined by the ECMS database.

Driving pattern recognition based on LVQ

The driving cycle plays an important role in PHEV control strategy. In order to have the driving cycle considered, a driving cycle recognition system based on LVQ is developed to support the application of the optimal EF map.

Feature parameters extraction

According to the Won and Langari,³³ the 11 selected parameters are as follows: the average speed $\bar{v}_g$ calculated by equation (24), the maximum speed $v_{g\max}$ calculated by equation (25), the maximum acceleration $a_{g\max}$ calculated by equation (26), the average acceleration $\bar{a}_g$, the max deceleration $d_{g\max}$ calculated by equation (27), the average deceleration $\bar{d}_g$ calculated by equation (28). The percent of idling time in total cycle time r_i , the percent of accelerating time in total cycle time r_a , the percent of decelerating in total cycle time r_d , the percent of constant velocity time in total cycle time r_c and the times of idle.

$$\bar{v}_g = \frac{\sum_{i=1}^{t_g} v_g}{t_g} \quad (24)$$

Where v_g denotes the instant vehicle velocity in cycle, t_g is the time of total cycle.

$$v_{g\max} = \max(v_g) \quad (25)$$

$$a_{g\max} = \max(a_g) \quad (26)$$

Where a_g denotes the instant acceleration in the cycle.

$$d_{g\max} = \max(d_g) \quad (27)$$

Where d_g denotes the instant deceleration in the cycle.

$$\bar{d}_g = \frac{\sum_{i=1}^{t_g} d_g}{t_g} \quad (28)$$

Establishing the LVQ neural network and simulation

The algorithm of the LVQ neural network is considered as an efficient method to recognize a complex and non-linear object.³⁴ In this account, LVQ is introduced to develop the driving pattern recognition system. In this paper, 11 feature parameters corresponding to 11 input neurons and three output neurons corresponding to three types of different patterns (1-NYCC stop and go pattern, 2-UDDS urban and suburban pattern, 3-HWEFT high speed pattern) are extracted to constitute the database for estimating the driving cycle. The LVQ network is trained with the database. Figure 12 shows the recognition training result, with the training times increasing, the error rate decreasing and gradually approaching 0 in the 54th training. A series of prior test data is used to test the network. As Figure 13 shows, the test results errors are close to 0, the method can recognize the driving cycle well.

Adaptive control strategy based on EF optimal map combined with DPR

A-ECMS Energy Management strategy is established based on an optimal map of EF combined with driving

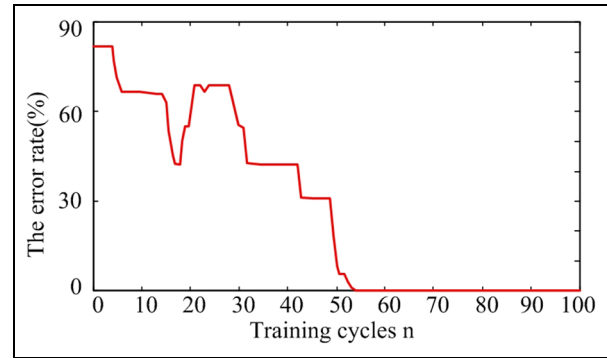


Figure 12. The error of recognition during training, the error closes to zero after 60 generations.

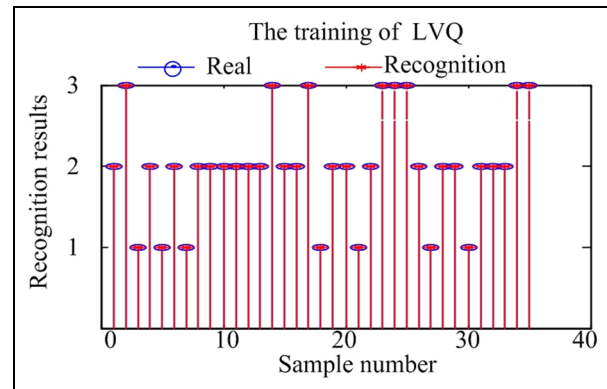


Figure 13. The final recognition error of the test cycle.

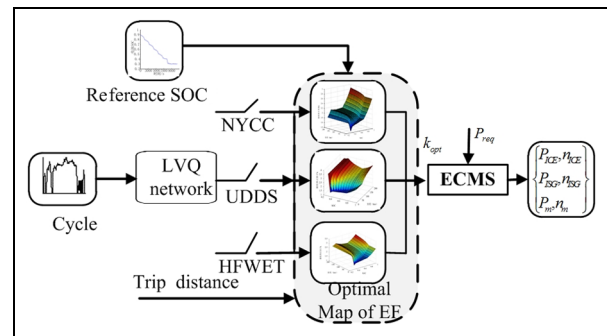


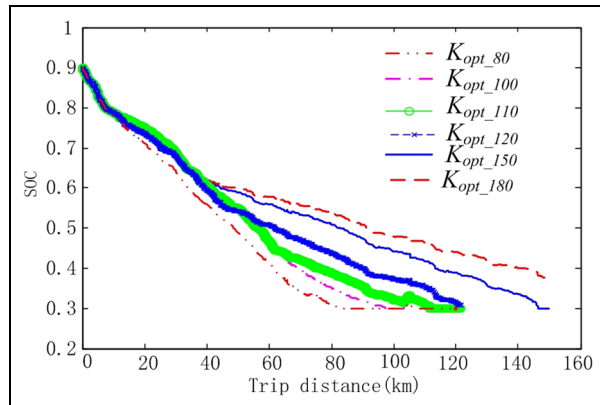
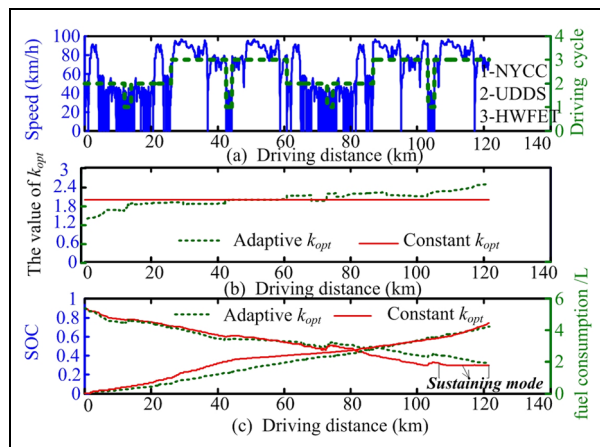
Figure 14. Schematic of the adaptive control strategy based on EF optimal map combined with DPR.

pattern recognition, which consists of the LVQ network and the optimal correction EF map. The schematic of the strategy is shown in Figure 14.

In this research, a map-based approach combined with DPR is introduced to resolve the question of how to adjust the EF corresponding to the real-world driving environment, as a section of designing the adaptation laws for real-time ECMS when the battery real-time SOC violates the reference SOC. This method consists of two sections, one is the current driving pattern identification, and another is the expectation of trip distance adaptation. For example, when the vehicle is

Table 2. Basic parameters of the vehicle.

Symbol	Parameter	Value
A	Frontal area	2.13 m ²
m_{veh}	Gross mass	1500 kg
i_0	Final ratio	5.297
C_d	Air drag coefficient	0.29
r_w	Tire radius	0.312

**Figure 15.** Test result of SOC trajectory for various prescribed distances.**Figure 16.** The results of DPR, correction factor k , SOC and fuel consumption: (a) The result of driving pattern Recognition; (b) The updating trajectory of correction factor k ; (c) The comparison results of SOC and fuel consumption.

driving, the current driving cycle will be identified by the module of the DPR and the corresponding map of optimal EF will be applied to properly adjust the EF to acquire an optimal performance under the trip distance of the destination scheduled.

Validation results and discussion

With the help of Matlab/Simulink software, a forward PHEV model was developed to examine the optimal adaption EF of the control strategy. Two case studies, a

different driving distances adaptive test and a driving pattern adaptive test, have been demonstrated to show the sensitive effect of EF adjustments. A HIL experiment was conducted to validate the control strategy further. The basic parameters of the vehicle are listed in Table 2.

Trip distance adaptive test and result

For PHEV, the trip distance adaptation is one of the most important parts to the control strategy. In order to examine the trip distance adaptation of the proposed strategy based on the EF, it was designed at various EFs for different prescribed distances, such as 80, 100, 110, 120 km and so on. The test results are shown in Figure 15, the SOC trajectory of various distances are different from each other, and it indicates that the proposed strategy based EF can be adapted perfectly when the expectation of travel distance is changing.

Driving pattern adaptive test and result

Figure 16(a) shows a test driving cycle which consists of six representative cycles executed by the PHEV model. The LVQ network can recognize the driving cycle well. To examine the improved potential fuel economy performance of the proposed strategy, a conventional ECMS with constant optimal penalty factor K_{opt} is considered as the baseline strategy.

In Figure 16(b), the red line represents the baseline strategy for which the penalty factor K_{opt} is equal to 1.891, the green dashed line denotes the recognition strategy penalty factor K which is tuned for different cycles. The correction factor increases with increasing driving distance for the same driving cycle, which explains why the EF is larger and the battery power is consumed more slowly with increasing driving distance.

In Figure 16(c), the SOC value decreases more rapidly under the baseline strategy, when the distance is 105 km, the SOC comes to 0.3, and the vehicle switches into CS mode since the SOC can maintain the balance around 0.3, as shown in Figure 16(c) noted "Sustaining mode". The adaptive strategy works in the blend mode in the whole cycle and the SOC comes to 0.3 at the end of the cycle. In addition, in the adaptive strategy the equivalent fuel consumed is 4.153 L, while in the constant strategy it is 4.775 L. The fuel consumption is decreased by 13.03%.

The working area of the engine under adaptive strategy and constant strategy is shown in Figure 17. Compared to the constant strategy, the engine working points of the adaptive strategy tend to be located in the higher efficiency area. This result indicates that the proposed strategy is able to acquire the relative minimum fuel consumption performance.

Another case was studied to further validate and evaluate the effectiveness of the proposed control strategy. In this case, a constant EF of 2.6 was used to compare with the adaptive EF. The results are shown in

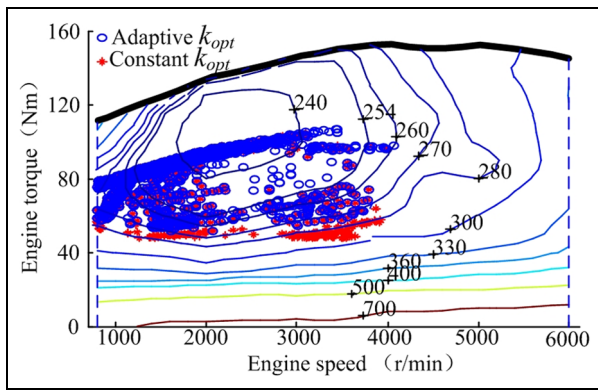


Figure 17. Comparison of engine performance result.

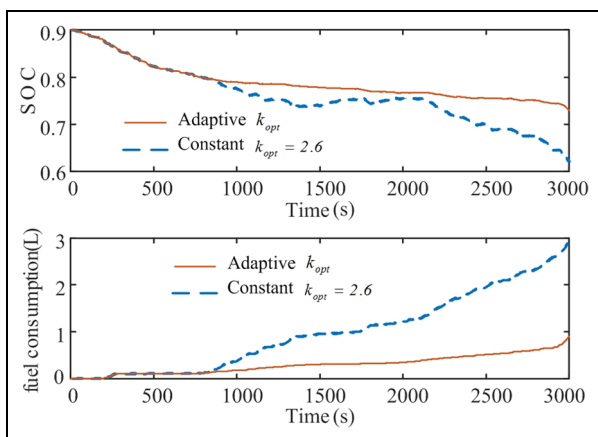


Figure 18. The comparison results of SOC and fuel consumption.

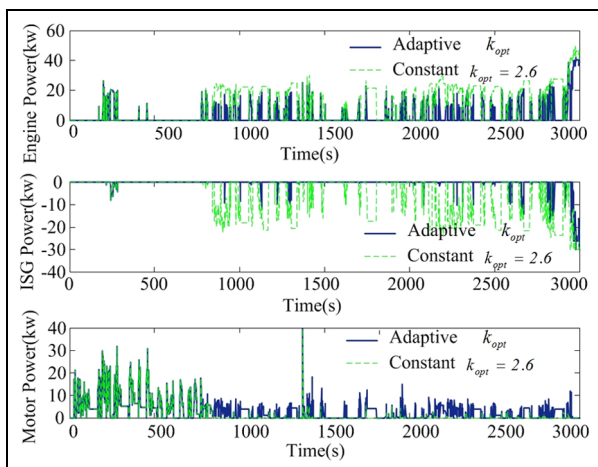


Figure 19. The power distribution comparison of the two strategies.

Figures 18 and 19. From Figure 18 it can be seen that the adaptive EF results in a smoother SOC trend than in the constant EF. The proposed strategy acquires a better fuel economy performance than the constant EF strategy.

Figure 19 shows the results of the power distribution simulation. From the results of the power distribution comparison of the two strategies, it can be clearly observed that the constant EF strategy tends to use more engine power transferred to ISG power for charging the battery, since the EF cannot update according to the driving cycle. The power distribution is less unreasonable in the constant EF strategy. On the contrary, in the adaptation EF strategy with the capability to update the EF by DPR, both the SOC and fuel consumption performance are better. In this case, it suggests that the adaptation EF is able to improve the fuel economy of PHEV.

Hardware in loop experiment bench

The HIL experiment was conducted to evaluate the performance of the proposed energy management strategy. The HIL experiment bench is shown in Figure 20; there is one pair of electric drive systems, which include a battery pack, battery management, dual motors and their controllers. In the HIL system, one motor serves as the traction power and the other as the power load. The process of HIL is expressed as follows: the D2P is the vehicle controller to control the electric drive system and battery management system. The PHEV simulation model, based on Simulink, can be transfer into the control code and downloaded into the D2P. The load controller station serves as the driving operation platform to be manipulated by the tester. The monitor system will observe the state of the hardware and log the essential data.

The HIL test results are compared to those of the simulation. The trend of EF, SOC and fuel as shown in Figure 21. The test result is denoted by a solid line, the simulation is the dashed line.

It is clearly observed that there is a little deviation between the HIL test and simulation. Fortunately, the SOC and fuel trends of the HIL test are globally consistent with that of the simulation. From the test results we can obtain the information that the optimal final SOC, as well as the optimal fuel consumption, can be achieved by employing the adaptation EF control strategy.

Conclusions

1. This research proposes an optimal adaptation EF energy management strategy for a plug-in CVT HEV based on ECMS combined with DPR. The EF optimization model, as a function of the SOC and trip distance, was developed based on the ECMS. The EF global optimization maps, as the function of trip distances and SOC in various driving cycles, are extracted from the GA. LVQ was introduced for the identification of the driving mode and to achieve a properly optimal EF update in real-time in the corresponding driving cycle.
2. Both the driving distances adaptive test and driving pattern adaptive test were studied to validate the

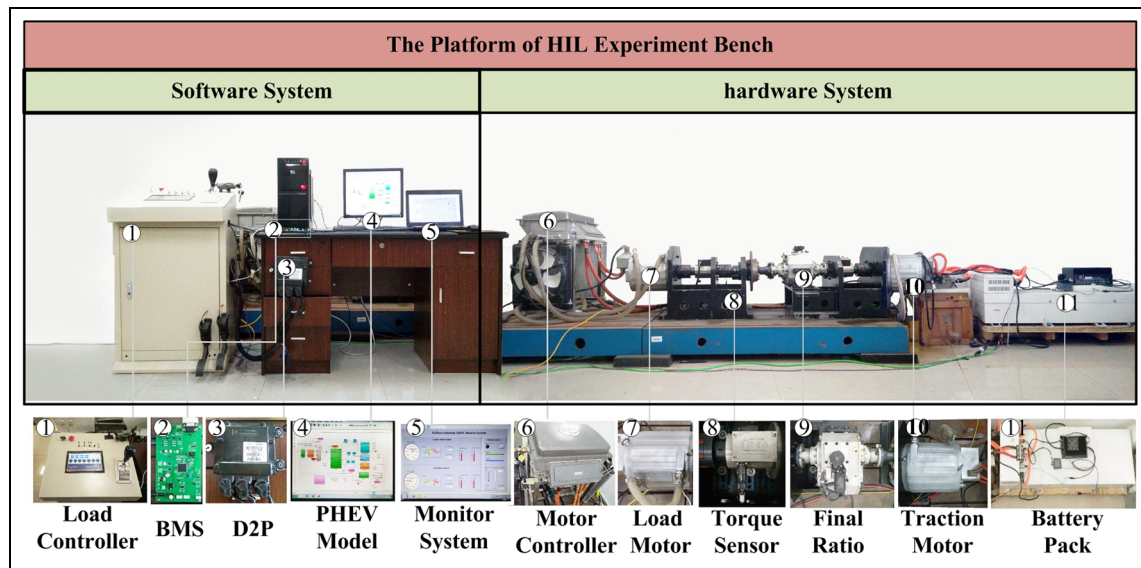


Figure 20. The platform of the HIL experiment bench.

1 Load Controller, 2 Battery management system, 3 D2P HIL ECU, 4 Simulink model, 5 Data collection and monitor system, 6 Motor controller, 7 Load motor, 8 Torque sensor, 9 Final drive ratio, 10 Traction motor, 11 Lithium ion battery pack.

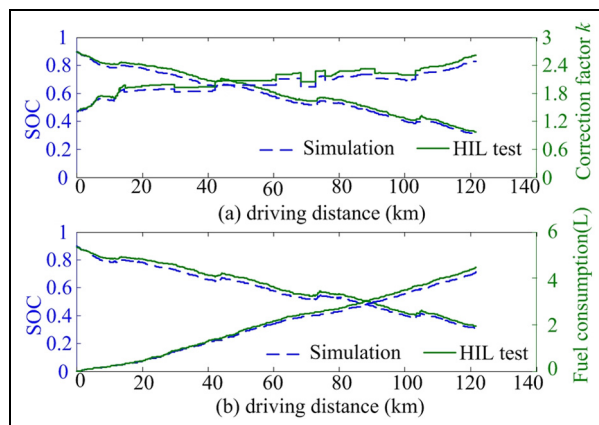


Figure 21. Comparison of fuel economy for simulation and HIL.

effectiveness of the proposed control strategy. The test results indicate that the proposed strategy will be able to adjust the EF online in the corresponding driving cycle, and the fuel economy is improved by an increase of 13.03% compared with the constant EF strategy.

3. A HIL experiment was also conducted to evaluate the performance of the proposed strategy. The HIL test results indicate that the proposed optimal adaptation EF energy management strategy can adjust the power distribution properly in real-time according to the driving cycle and is capable of further improving the fuel economy for PHEV.

Acknowledgements

The authors greatly appreciate the support of the China Scholarship Council to support the author as a visiting scholar in WVU. This work was completed at the

Center for Alternative Fuels, Engines and Emissions of the West Virginia University.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work has been financial supported by the National Natural Science Foundation of China (grant number 51505086).

References

1. Manzie C, Dewangan P, Corde G, et al. State of charge management for plug in hybrid electric vehicles with uncertain distance to recharge. In: *9th Asian control conference (ASCC)*, Istanbul, 2013, pp.1–6.
2. Wirasingha SG and Emadi A. Classification and review of control strategies for plug-in hybrid electric vehicles. *IEEE Trans Veh Technol* 2011; 60(1): 111–122.
3. Gong QM, Li YY and Peng ZR. Trip based optimal power management of plug-in hybrid electric vehicles using gas-kinetic traffic flow model. In: *2008 American control conference*, Seattle, WA, 2008, pp.3225–3230.
4. Yu H, Kuang M and McGee R. Trip-oriented energy management control strategy for plug-in hybrid electric vehicles. *IEEE Trans Control Syst Technol* 2014; 22(4): 1323–1336.
5. Montazeri-Gh M and Mahmoodi-K M. Development a new power management strategy for power split hybrid electric vehicles. *Transp Res Part D: Transp Environ* 2015; 37: 79–96.

6. Zhang S and Xiong R. Adaptive energy management of a plug-in hybrid electric vehicle based on driving pattern recognition and dynamic programming. *Appl Energy* 2015; 155: 68–78.
7. Li L, Yang C, Zhang Y, et al. Correctional DP-based energy management strategy of plug-in hybrid electric bus for city-bus route. *IEEE Trans Veh Technol* 2015; 64(7): 2792–2803.
8. Delprat S, Lauber J, Guerra TM, et al. Control of a parallel hybrid powertrain: optimal control. *IEEE Trans Veh Technol* 2004; 53(3): 872–881.
9. Larsson V, Johansson L and Egardt B. Analytic solutions to the dynamic programming subproblem in hybrid vehicle energy management. *IEEE Trans Veh Technol* 2015; 64(4): 1458–1467.
10. Peng JK, He HW and Xiong R. Rule based energy management strategy for a series-parallel plug-in hybrid electric bus optimized by dynamic programming. *Applied Energy* 2017; 185(2): 1633–1643.
11. Hegazy O, Mierlo JV and Lataire P. Design optimization and optimal power control of fuel cell hybrid electric vehicles based on swarm intelligence. *Int Rev Electr Eng* 2011; 6(4): 1727–1738.
12. Wu L, Wang Y, Yuan X, et al. Multiobjective optimization of HEV fuel economy and emissions using the self-adaptive differential evolution algorithm. *IEEE Trans Veh Technol* 2011; 60(6): 2458–2470.
13. Geng B, Mills JK and Sun D. Energy management control of microturbine-powered plug-in hybrid electric vehicles using the telemetry equivalent consumption minimization strategy. *IEEE Trans Veh Technol* 2011; 60(9): 4238–4248.
14. Serrao L, Onori S and Rizzoni G. ECMS as a realization of Pontryagin's minimum principle for HEV control. In: *2009 American control conference*, St. Louis, MO, 2009, pp.3964–3969.
15. Borhan H, Vahidi A, Phillips AM, et al. MPC-based energy management of a power-split hybrid electric vehicle. *IEEE Trans Control Syst Technol* 2012; 20(3): 593–603.
16. Camacho DEF and Bordons DC. *Model predictive control*. vol. 2. London: Springer-Verlag, 2004, pp.347–365.
17. Zhao H and Wu D. A research on the control strategy of a parallel HEV based on stochastic model predictive control. *Automot Eng* 2014; 36(11): 1289–1294.
18. Zhang P, Yan F and Du C. A comprehensive analysis of energy management strategies for hybrid electric vehicles based on bibliometrics. *Renew Sustain Energy Rev* 2015; 48: 88–104.
19. Lin XY and Sun DY. Driving pattern recognition based on ECMS and its application to control strategy for a series-parallel hybrid electric bus. *J Hunan Univ* 2012; 39(10): 43–49.
20. Sivertsson M and Eriksson L. Design and evaluation of energy management using map-based ECMS for the PHEV benchmark. *Oil Gas Sci Technol* 2014; 70(1): 195–211.
21. Chen HH, Kessels JJ and Weiland SS. Adaptive ECMS: a causal set-theoretic method for equivalence factor estimation. *IFAC Pap OnLine* 2015; 48(15): 78–85.
22. Onori S, Serrao L and Rizzoni G. Equivalent consumption minimization strategy. In: *Hybrid electric vehicle energy management strategies*. London: Springer-Verlag, 2016, pp.65–76.
23. Onori S and Serrao L. On adaptive-ECMS strategies for hybrid electric vehicles. In: *Proceedings of the international scientific conference on hybrid and electric vehicles*, Malmaison, France, 2011, pp.6–7.
24. Tianheng F, Lin Y, Qing G, et al. A supervisory control strategy for plug-in hybrid electric vehicles based on energy demand prediction and route preview. *IEEE Trans Veh Technol* 2015; 64(5): 1691–1700.
25. Vajedi M, Chehrehshaz M and Azad NL. Intelligent power management of plug-in hybrid electric vehicles, part ii: real-time route based power management. *Int J Electr Hybrid Veh* 2014; 6(1): 68–86.
26. Zhang C and Vahidi A. Route preview in energy management of plug-in hybrid vehicles. *IEEE Trans Control Syst Technol* 2012; 20(2): 546–553.
27. Seelan V. Analysis, design and application of continuously variable transmission (CVT). *Int J Eng Res Appl* 2015; 5(3): 99–105.
28. Sun D, Lin X, Qin D, et al. Power-balancing instantaneous optimization energy management for a novel series-parallel hybrid electric bus. *Chin J Mech Eng* 2012; 25(6): 1161–1170.
29. Sciarretta A, Back M and Guzzella L. Optimal control of parallel hybrid electric vehicles. *IEEE Trans Control Syst Technol* 2004; 12(3): 352–363.
30. Pei D and Leamy MJ. Dynamic programming-informed equivalent cost minimization control strategies for hybrid-electric vehicles. *J Dyn Syst Meas Control* 2013; 135(5): 051013.
31. Paganelli G, Ercole G, Brahma A, et al. General supervisory control policy for the energy optimization of charge-sustaining hybrid electric vehicles. *JSAE Rev* 2001; 22(4): 511–518.
32. Tulpule PL. Energy management for plug-in hybrid electric vehicles using equivalent consumption minimisation strategy. *Int J Electr Hybrid Veh* 2010; 2(4): 329–350.
33. Won JS and Langari R. Intelligent energy management agent for a parallel hybrid vehicle. In: *Proceedings of the 2003 American control conference*, 2003, vol. 3, pp.2560–2565.
34. Sundström O, Ambühl D and Guzzella L. On implementation of dynamic programming for optimal control problems with final state constraints. *Oil Gas Sci Technol* 2009; 65(1): 91–102.

Appendix I

Notation

m_v	the mass of the vehicle
v	the longitudinal vehicle velocity
F_{out}	the force generated by the powertrain
F_{bra}	the braking force of the vehicle
θ	the road gradient
r_w	the radius of the wheel
C_d	air drag coefficient
A	the frontal area of the vehicle
g	the gravitational acceleration
f_r	the rolling resistance of the road
δ	the equivalent inertia factor
i_0	the final drive ratio
i_{cvt}	the CVT ratio

η_t	the efficiency of the transmission (including driving axles, CVT, final reducer and differential)	m_{eng}	the engine fuel consumption during the whole trip
P_{eng}	engine power	Q_{lHV}	the lower heating value (LHV) of the diesel engine fuel (43,000 KJ/kg)
P_{EM}	traction motor power	Q_0	the capacity of the battery
P_{ISG}	ISG power	I_{batt}	the current of the battery
P_{bra}	braking power	V_{oc}	the open circuit voltage of the battery
t_0	the trip initial time	R_{batt}	internal resistance of the battery
t_f	the trip terminal time	P_{batt}	the battery power
b_e	the fuel consumption rate	η_{cvt}	the efficiency of the CVT
T_{eng}	the engine torque	i_{cvt}	the gear ratio
ω_{eng}	engine speed	T_{cvt}	the input torque of the CVT
$\dot{m}_{eng}$	the instantaneous fuel consumption rate of the engine		